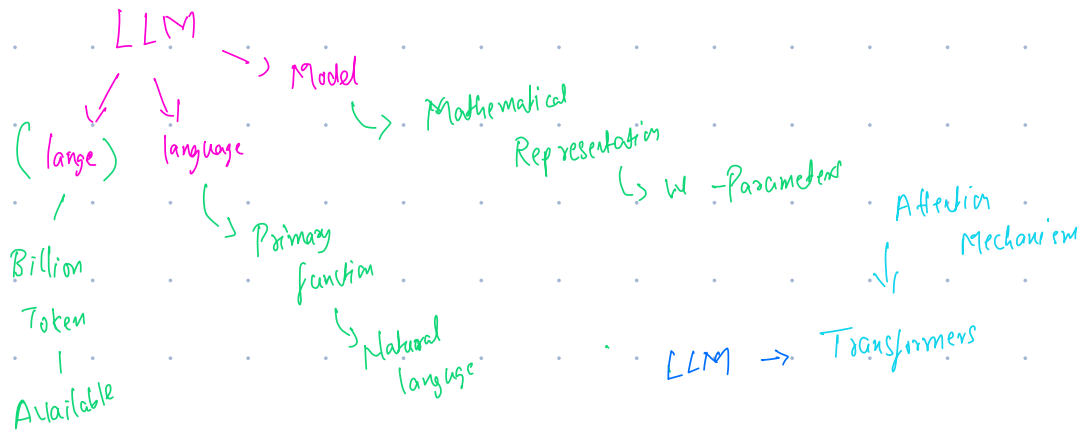
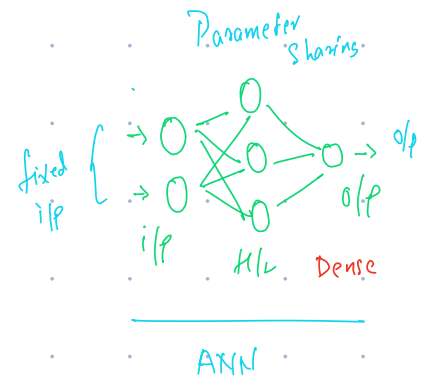
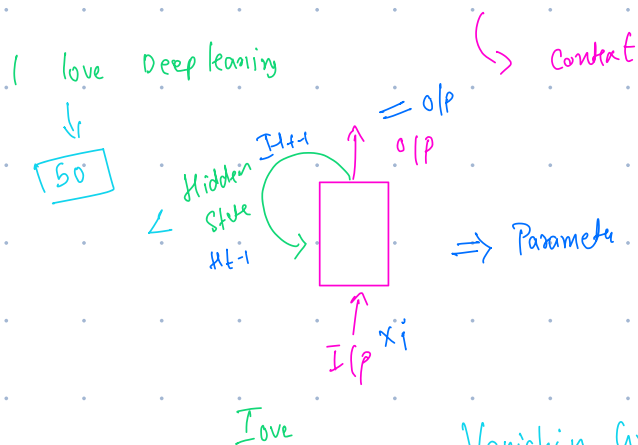




GenAI - ChatGPT, Gemini - LLM



RNN - Recurrent Neural Network



Sequential Data

$\rightarrow$ Text, Speech

Varying length $\rightarrow$ Time Series

$\rightarrow$ 10 - 15 - 30

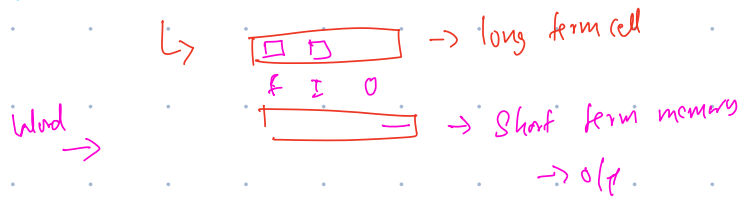
Vanishing Gradient ✓

Explosion Gradient ✓

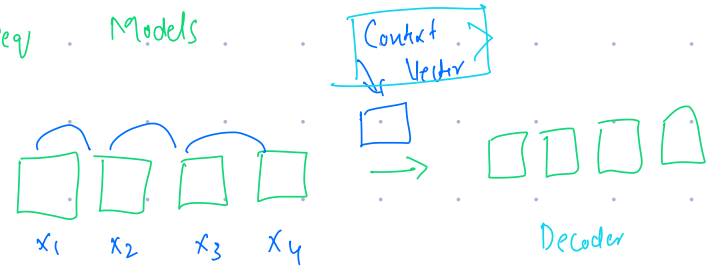
$$0.99^{50} = 0.0099 \rightarrow \text{Zero}$$

$$1.01^{50} = 1. \rightarrow \text{larger}$$

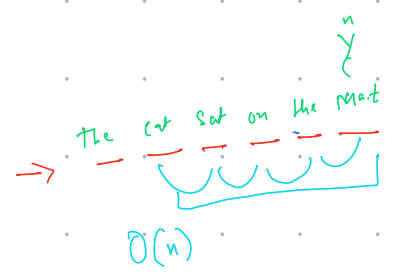
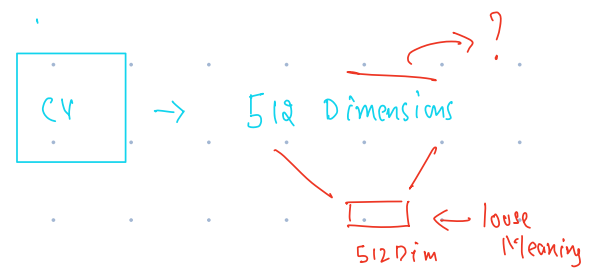
LSTM - long short term Memory



Seq2Seq Models



1000 kinds



The long Range Dependency

2017 - "The Attention is All You Need"

Attention

Transformers

Vaswani

← Base

$O(n)$ - sequentially - RNN



Students - 3 (A, B, C)

① Student - self

A - A
B - B
C - C

② Look

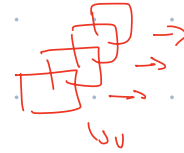
A - B, C
B - A, C

Attention Score $\rightarrow K, Q, V$
 $\swarrow \quad \searrow$
 Key Query Value

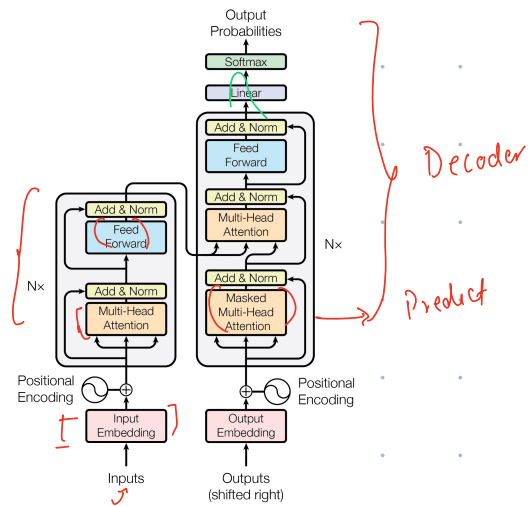
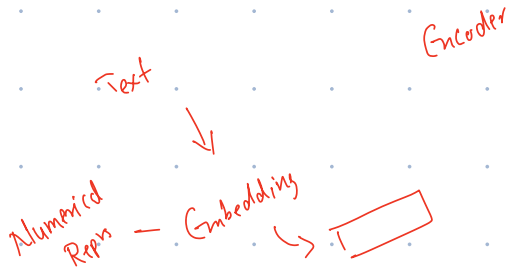
$$\begin{cases} Q = x \cdot W_Q \\ K = x \cdot W_K \\ V = x \cdot W_V \end{cases}$$

$$C = B \cdot A$$

③ $A = \sin y, B = c \cdot y_0$



Transformer $\rightarrow$ LLM



Encoder
 BERT
 CoBERT
 RoBERTa

Encoder-decoder
 (TG-flow)

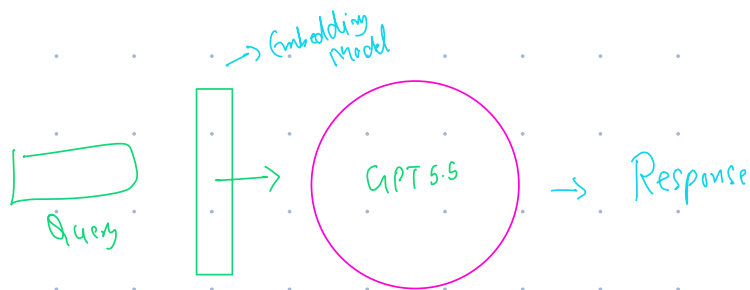
Decoder
 ChatGPT
 Gemini
 Claude
 Misha

Embeddings $\rightarrow$ Numerical Data

(King) - $[0.99, 0.66, 0.23]$
 $\downarrow$
 Word $\rightarrow$ Embedding Vector

Binary - 010101

OHE
 Label Encoder



Semantic Similarity

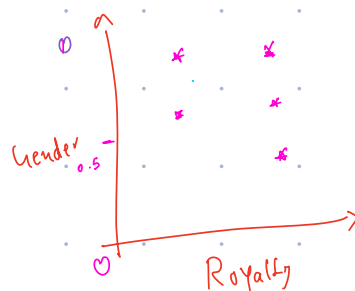
King [0.98, 0.96, 0.69]

3 Dimensions

f M
0 - 0.5 0.5 - 1
0.1, 0.2, 0.3, 0.4, 0.5

Context Semantic Understanding

	Gender	Royalty
King	[0.8	0.9]
Queen	[0.35	0.9]
Man	0.8	0.4
Woman	0.30	0.4
Boy	0.6	0.4
Princ	0.65	0.9



Text Embedding

large - (2018) Dim

3 Small

(1536)

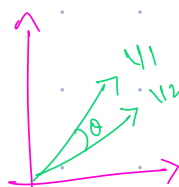
(384)

All mini ln 16 1/2

v1 - [1536]

v2 - [1536]

Cosine Similarity

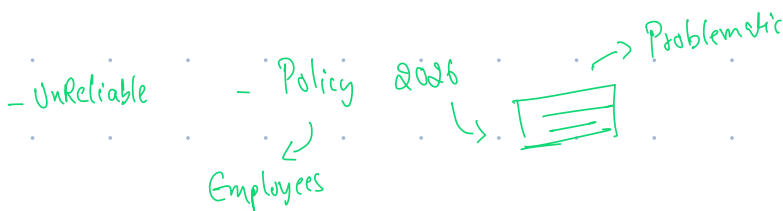
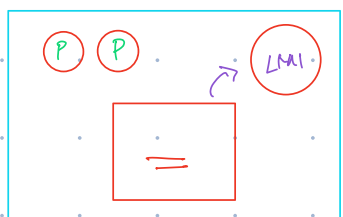
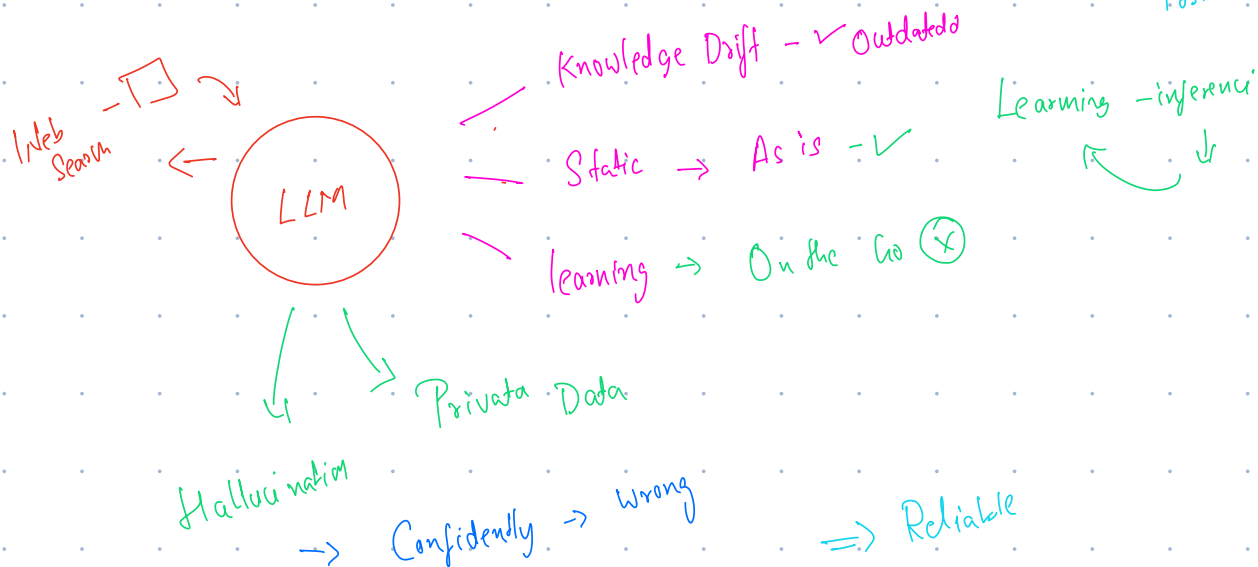
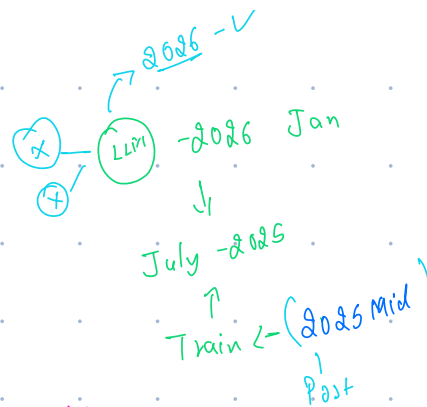
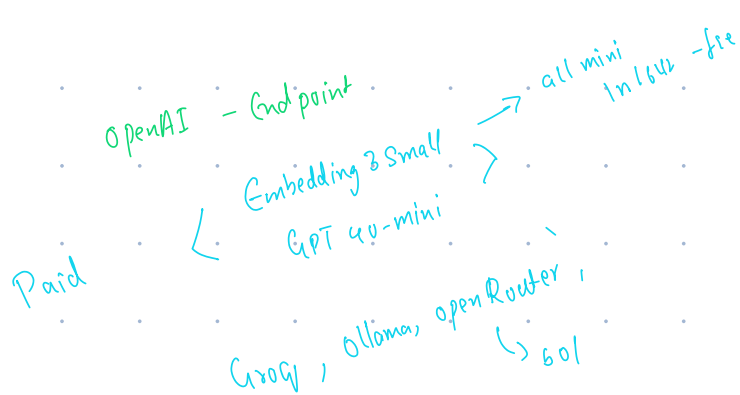


Missing Jan

Cosine = 1 - Similar

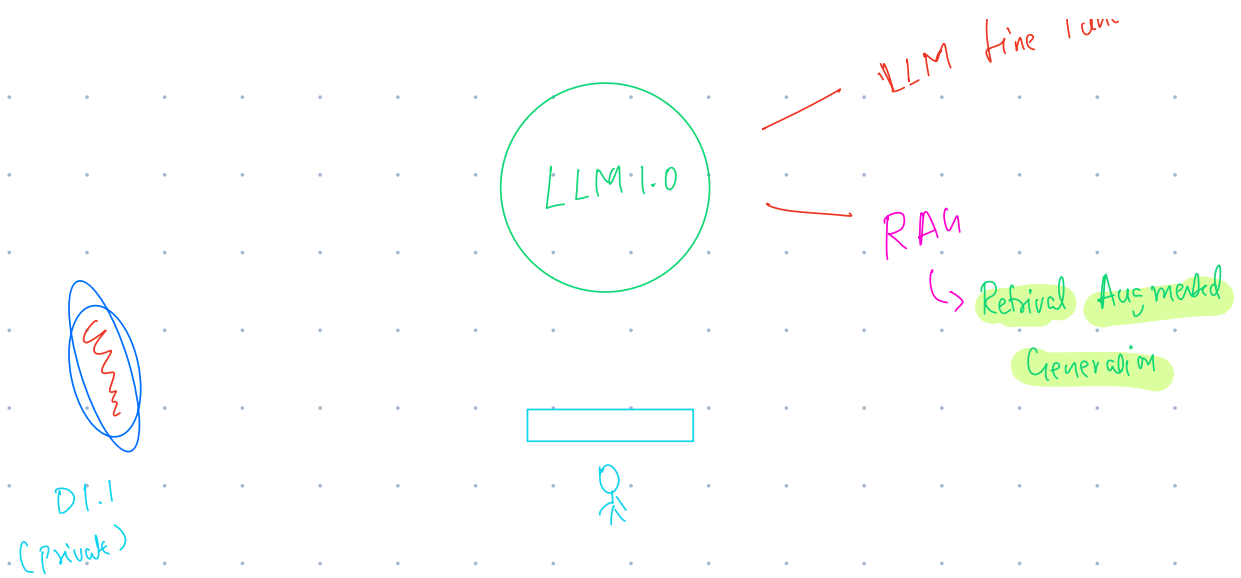
0 - Not

-1 = Diss/opp



Source - knowledge Base





Query $\rightarrow$ LLM $\rightarrow$ Response

Query $\rightarrow$ Retrieve $\rightarrow$ LLM $\rightarrow$ Response

Retrieve Generate

lookup

3 Stages

loading - Storing - Retrieval

- ① Data Ingestion
- ② Retrieval $\rightarrow$ Search $\rightarrow$ Right Info
- ③ Generation - Docs $\rightarrow$ LLM $\rightarrow$ Response

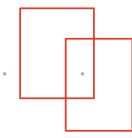
Data Ingestion $\rightarrow$

Fixed Sized = 800 / Totu

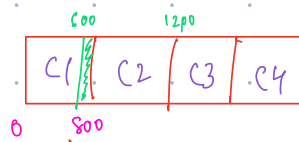
100

1536 Dimension

Numerical



Documents
Pdf, 100/pages

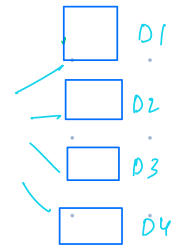


Chunking

Compressed

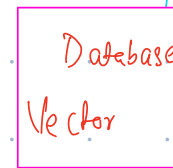


Embedding



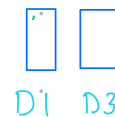
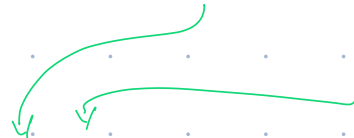
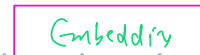
Qdrant
Zilliz

Chromadb
Pinecone
Milvus
faiss

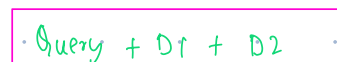


Cosine
Similarity

② Retrieval



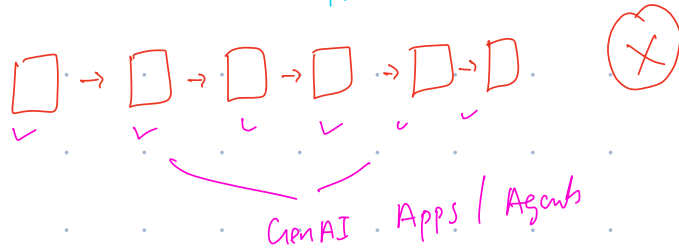
③ Generation



Response

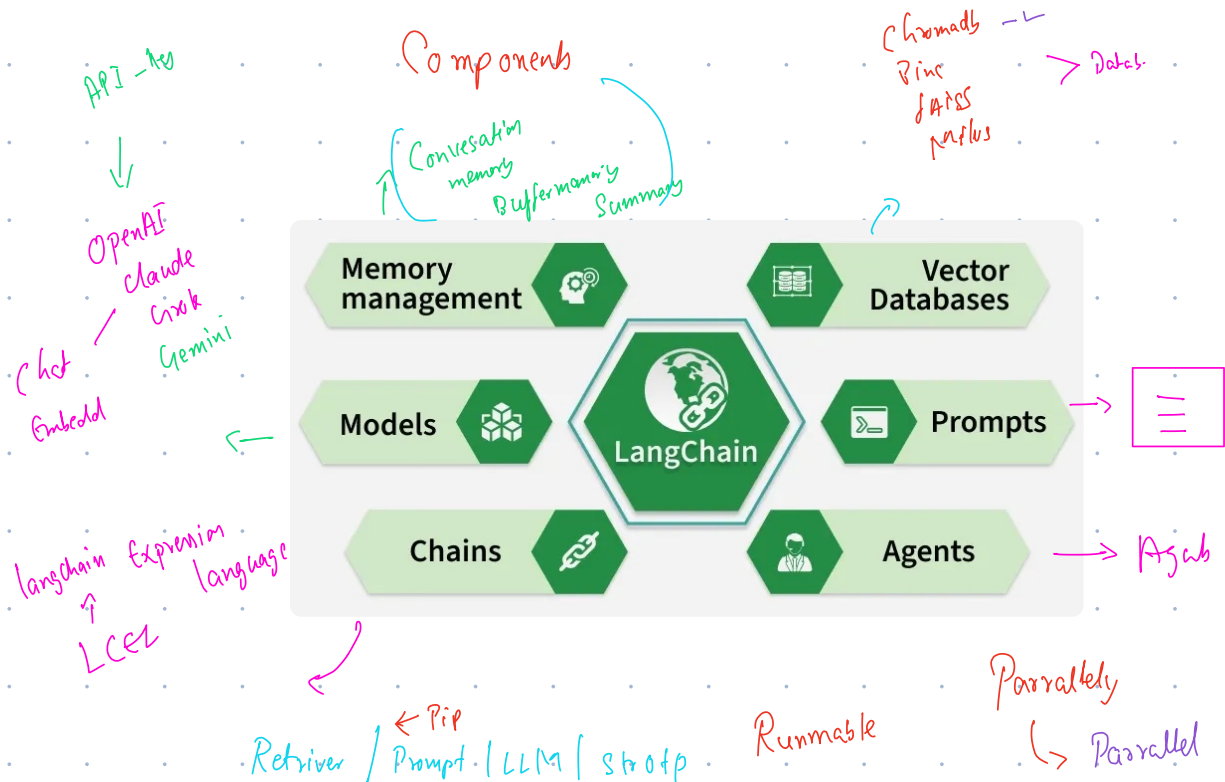
- ✓ - Source
- ✓ - Reliable
- ✓ - Document

Document loaders
 (chunking)
 (Database)
 Indexing - VDBs
 Prompt template
 Embedding
 LLM
 Prompt
 Reranking →



Langchain

→ frameworks → Thing
 Modules
 Composable



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{ $\equiv$ }

Code